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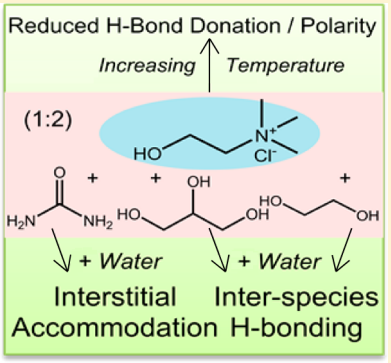
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Solvatochromic Probe Behavior within Choline Chloride-Based Deep Eutectic Solvents: Effect of Temperature and Water

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ABSTRACT: Deep eutectic solvents (DESs) have shown potential as promising environmentally friendly alternatives to conventional solvents. Many common and popular DESs are obtained by simply mixing a salt and a H-bond donor. Properties of such a DES depend on its constituents. Change in temperature and addition of water, a benign cosolvent, can change the physicochemical properties of DESs. The effect of changing temperature and addition of water on solvatochromic probe behavior within three DESs formed from choline chloride combined with 1,2-ethanediol, glycerol, and urea, respectively, in 1:2 mol ratios termed ethaline, glyceline, and reline is presented. Increase in temperature results in reduced H-bond donating acidity of the DESs. Dipolarity/polarizability and H-bond accepting basicity do not change with changing temperature of the DESs. The response of the fluorescence probe pyrene also indicates a decrease in the polarity of the DESs as temperature is increased. Addition of water to DES results in increased dipolarity/polarizability and a decrease in H-bond accepting basicity. Except for pyrene, solvatochromic probes exhibit responses close to those predicted from ideal-additive behavior with slight preferential solvation by DES within the aqueous mixtures. Pyrene response reveals significant preferential solvation by DES and/or the presence of solvent−solvent interactions, especially within aqueous mixtures of ethaline and glyceline, the DESs constituted of H-bond donors with hydroxyl functionalities. FTIR absorbance and Raman spectroscopic measurements of aqueous DES mixtures support the outcomes from solvatochromic probe responses. Aqueous mixtures of ethaline and glyceline possess relatively more interspecies H-bonds as compared to aqueous mixtures of reline, where interstitial accommodation of water within the reline molecular network appears to dominate.



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INTRODUCTION

Having shown significant advantages over conventional ionic liquids, especially in terms of toxicity, cost, and ease of preparation/handling, deep eutectic solvents (DESs) are in the process of establishing themselves as one of the premier choices of solvents in science and technology. DESs, which are also called ionic liquid analogues, are obtained by simply mixing two (or more) appropriate compounds, most commonly a salt and a H-bond donor (HBD), followed by gentle heating. The components that form common and popular DESs are usually cheap and nontoxic materials, e.g., salt choline (2-hydroxyethyltrimethylammonium) chloride (this is vitamin B_4) among several ammonium and phosphonium salts, and urea, ethylene glycol, and glycerol among several HBDs.[1,2] DESs are academically interesting due both to their inherent structural complexities and to their enormous potential as solvents in organic catalysis,[3] electrochemistry,[3] biochemistry, and a variety of other applications.[3−8] DESs are documented to be fairly nontoxic[9] and are capable of dissolving several classes of solutes.[1−10]

As expected, the properties of a DES depend on its constituents, the salt and the H-bond donor. The interactions present between the constituents of the DES usually govern its physicochemical properties. External means, such as a change in temperature and addition of a cosolvent, respectively, may significantly change the key properties of a DES. Variation in temperature within a DES, in this context, would not only reveal the changes in the key physicochemical properties of the DES, but it would also reveal the nature of the interactions present within the system. Although, as mentioned earlier, the properties of a DES could be specific to the salt−HBD combination, a mixture of a DES with other solvents may afford improved and favorable physicochemical properties. Aqueous mixtures of DESs, in this respect, have garnered increased attention.[11−19] The major reason for this could be traced to the proposition that the aqueous mixtures of DESs may form a class of "hybrid green" system. The fact that several common DESs are not only hygroscopic in nature but they exhibit complete miscibility with water further contributes to the need and growing interest to understand aqueous DES mixtures. Due to the possibility of strong intermolecular H-bonding interactions, among others, between water and the constituents of a DES, addition of water may potentially change the physicochemical properties of DESs in a significant fashion. In this regard, though the investigation of structural features of the solution as well as measurement of bulk physical properties of aqueous mixtures of DESs are of certain importance,[11−19] understanding the behavior of solutes dissolved in this "hybrid green" media may directly furnish crucial information on solute−solvent interaction(s). Key insights on physicochemical

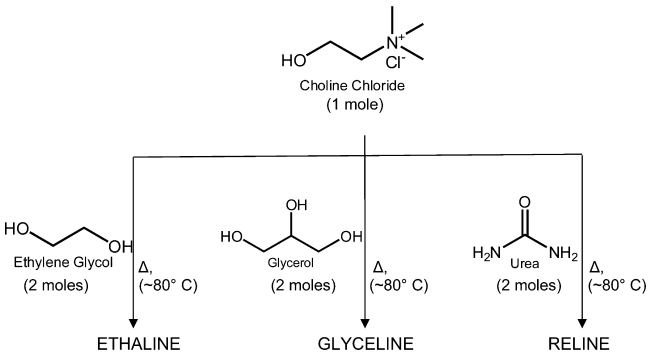
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properties of aqueous mixtures of DESs along with information on solute solvation within such systems would be obtained in the process.

In this paper, we present results of our investigation on three common and popular choline chloride-based DESs, ethaline, glyceline, and reline, prepared by mixing 1 mol of choline chloride with 2 mol of H-bond donor—1,2-ethanediol, glycerol, and urea, respectively (Scheme 1). In order to assess the effect of temperature and added water, respectively, on important physicochemical properties of the three DESs, we have chosen to observe the behavior of different solvatochromic absorbance and fluorescence probes as a function of temperature in the range 30−90 °C and as water is added to the DES. Specifically, we have used three common electronic absorbance probes, betaine dye 33, *N*,*N*-diethyl-4-nitroaniline (DENA), and 4-nitroaniline (NA), and three popular fluorescence probes, pyrene (Py), 6-propionyl-2-(dimethylaminonaphthalene) (PRODAN), and 1-anilino-8-naphthalenesulfonate (ANS), for this purpose (the structures of the probes are provided in Scheme 2). While the behavior of absorbance probes furnishes information on empirical parameters of importance for a medium, the three fluorescence probes are common empirical polarity probes. We also present outcomes of noninvasive FTIR absorbance and Raman spectroscopic investigations that corroborate results of solvatochromic probe responses within aqueous DES mixtures.

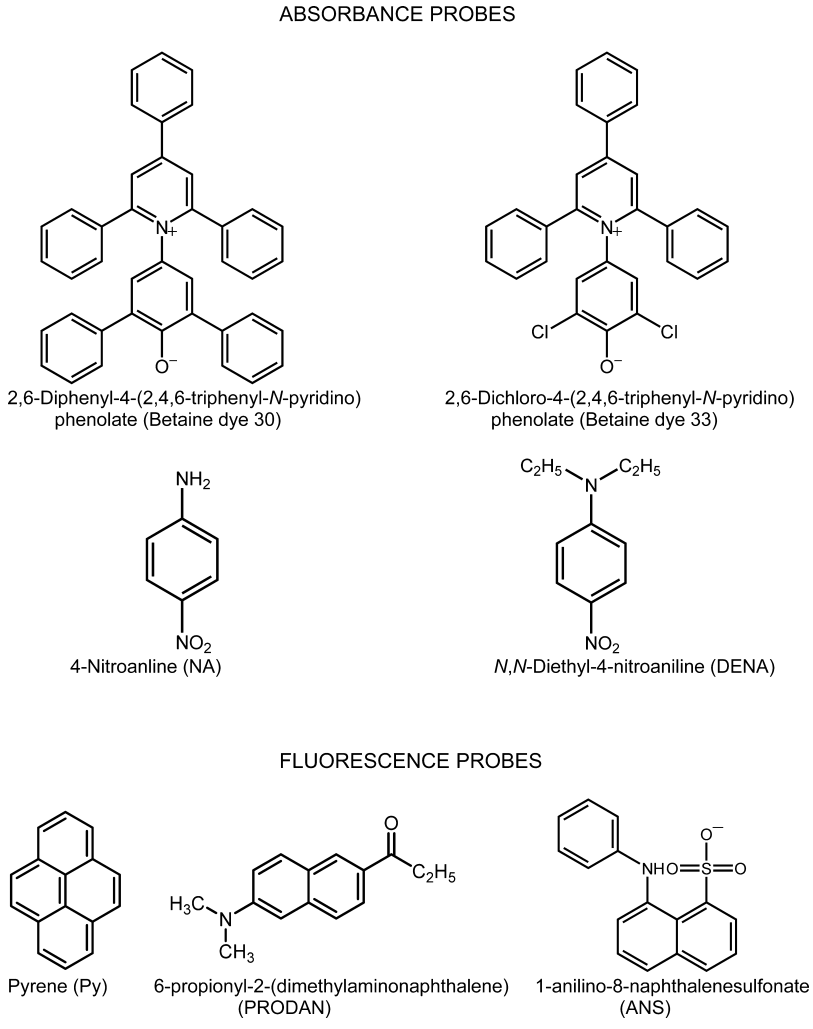
Scheme 1. DESs Used in This Study



EXPERIMENTAL SECTION

Materials. 2,6-Dichloro-4-(2,4,6-triphenyl-*N*-pyridino)-phenolate (betaine dye 33), 4-nitroaniline, and *N*,*N*-diethyl-4-nitroaniline were purchased in the highest available purity from Fluka (≥99%, HPLC), Spectrochem. Co. Ltd., and Frinton Laboratories, respectively. Pyrene [≥99.0% (GC), *puriss* for

Scheme 2. Molecular Structures of the Solvatochromic Probes Used



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fluorescence], 1-anilino-8-naphthalenesulfonate (99%), and 6-propionyl-2-(dimethylaminonaphthalene) [≥98% (HPLC)] were obtained in the highest purities from Sigma-Aldrich, Acros Organics, and Biochemika, respectively, and were used as received. All three DESs were purchased in highest purity from Scionix Ltd. and were stored in an inert environment before their use. Alternatively, ethaline, glyceline, and reline were prepared by mixing choline chloride (≥99% from Sigma-Aldrich) with 1,2-ethanediol (99.8%, anhydrous from Sigma-Aldrich), glycerol (≥99.5%, spectrophotometric grade from Sigma-Aldrich), and urea (≥99% from Sigma-Aldrich), respectively, in a mole ratio of 1:2 followed by stirring under heating (~80 °C) until a homogeneous, colorless liquid has been formed. All spectroscopic measurements on DESs purchased from Scionix Ltd. and those prepared by mixing choline chloride with the corresponding H-bond donor were found to be statistically similar. Absolute ethanol was used to prepare probe stock solutions. Doubly distilled deionized water with ≥18.0 MΩ·cm resistivity was obtained from a Millipore Milli-Q Academic water purification system.

Methods. Stock solutions of all probes were prepared by dissolving in ethanol in precleaned amber glass vials and stored at 4 ± 1 °C. The required amount of probes was weighed using a Mettler-Toledo AB104-S balance with a precision of ±0.1 mg. An appropriate amount of the probe solution from the stock was transferred to the 1 cm^2 quartz cuvette. Ethanol was evaporated using a gentle stream of high purity nitrogen gas. A precalculated amount of the DES or the (DES + water) mixture is directly added to the cuvette, and the solution is thoroughly mixed. The solubility of a probe within a DES or (DES + water) mixture is checked using the linearity of the absorbance and/or the fluorescence intensity versus the concentration plot(s).

A PerkinElmer Lambda 35 double beam spectrophotometer with variable bandwidth and Peltier-temperature controller is used for acquisition of the UV−vis molecular absorbance data. Steady-state fluorescence spectra were acquired on a Jobin-Yvon Fluorolog-3 (model FL-3-11) modular spectrofluorometer equipped with a 450 W Xe arc lamp as the excitation source and single-grating monochromators as wavelength selection devices with a photomultiplier tube as the detector. The temperature was controlled with a Thermo NESLAB RTE7 circulating chiller bath having a stability of ±0.01 °C. All absorbance and fluorescence data were acquired using 1 cm^2 quartz cuvettes. Attenuated and reflectance-Fourier-transform infrared (ATR-FTIR) absorbance data were acquired from 4000 to 400 cm^{-1} on an Agilent Technologies Cary 660 ATR double-beam spectrophotometer. The liquid samples were evenly spread on KBr pellets to record the FTIR spectra. Raman spectra were acquired with 532 nm excitation using a model no. X/01/220 XploRA PLUS Confocal Raman spectrometer.

All spectroscopic measurements were performed at least in triplicate starting from sample preparation, and the results were averaged. All spectra were duly corrected by measuring the spectral responses from suitable blanks prior to data analysis and statistical treatment. All fluorescence probes used were found to have adequate fluorescence quantum yields within DESs under investigation.

RESULTS AND DISCUSSION

Effect of Temperature. The fact that temperature exerts a profound effect on the physicochemical properties of solutions is well-established.[20−22] The effect of temperature on the solvatochromic probe behavior within ionic liquids was investigated earlier by several researchers.[17,23−27] In one of the relevant reports, among others, the temperature dependent polarity of the ionic liquid 1-butyl-3-methylimidazolium hexafluorophosphate ([bmim][PF_6]) was studied by the Bright group.[27] The authors emphasized that the HBD strength of imidazolium cation was strongly temperature dependent but HBA abilities were weak functions of temperature and added water.[27] We present the effect of temperature on the responses of solvatochromic probes dissolved in the DESs ethaline, glyceline, and reline. When possible, we have compared our outcomes with those reported for common and popular ionic liquids.

Response of Betaine Dye. We have first assessed the effect of temperature on the response of betaine dye 33 (Scheme 2), an effective UV−vis molecular absorbance probe, when dissolved in the DESs ethaline, glyceline, and reline, respectively. Similar to the more popular 2,6-diphenyl-4-(2,4,6-triphenyl-*N*-pyridino)phenolate (betaine dye 30), betaine dye 33 is also known to exhibit an unusually high solvatochromic absorbance band shift as the nature of the cybotactic region is changed.[28,29] We have used betaine dye 33 in our studies due to solubility restrictions of betaine dye 30 in DESs as well as in water. There is a considerable charge transfer in betaine dyes from the phenolate to the pyridinium part of the zwitterionic molecule. Consequently, the solvatochromic probe behavior of betaine dyes is strongly affected by the HBD acidity of the solvent along with the dipolarity/polarizability; HBD solvents stabilize the ground state more than the excited state.[30] The molar transition energy of the betaine dyes is a convenient reflection of its lowest energy intramolecular charge-transfer absorbance band maxima (λ_{max}^{abs}) and is expressed in terms of kcal·mol^{-1} according to the expression $E_T(30) = 28591.5/\lambda_{max}^{abs}$ (nm). The lowest energy absorbance transition of this dye [i.e., $E_T(33)$] is calculated the same way $E_T(30)$ is calculated. The $E_T(33)$ can be converted to normalized E_T^N values using eqs 1 and 2.[31,32]

$$E_T(30) = 0.9953(\pm 0.0287)E_T(33) - 8.1132(\pm 1.6546) \quad (1)$$

$$R = 0.9926, \text{ standard error of estimate} = 0.8320, n = 20$$

$$E_T^N = \frac{[E_T(30)_{SOLVENT} - E_T(30)_{TMS}]}{[E_T(30)_{WATER} - E_T(30)_{TMS}]} = \frac{[E_T(30)_{SOLVENT} - 30.7]}{32.4} \quad (2)$$

Here, TMS stands for tetramethylsilane and $E_T(30)_{WATER}$ = 63.1 kcal·mol^{-1} and $E_T(30)_{TMS}$ = 30.7 kcal·mol^{-1} are experimentally observed values. E_T^N is easier to conceive, as it is dimensionless and varies between 0 for TMS (extreme nonpolar) and 1 for water (extreme polar).

Experimentally obtained E_T^N within ethaline, glyceline, and reline, respectively, in the temperature range 30−90 °C are shown in Figure 1. A cursory examination reveals that an increase in temperature results in a decrease in E_T^N within each of the DESs investigated. A decrease in E_T^N implies a decrease in the dipolarity/polarizability and/or the HBD acidity of the medium. In general, "polarity" is suggested to usually decrease with increasing temperature due in major part to the increased average thermal reorientation of the dipoles.[33] This results in a decrease in dielectric constant with increasing temperature of

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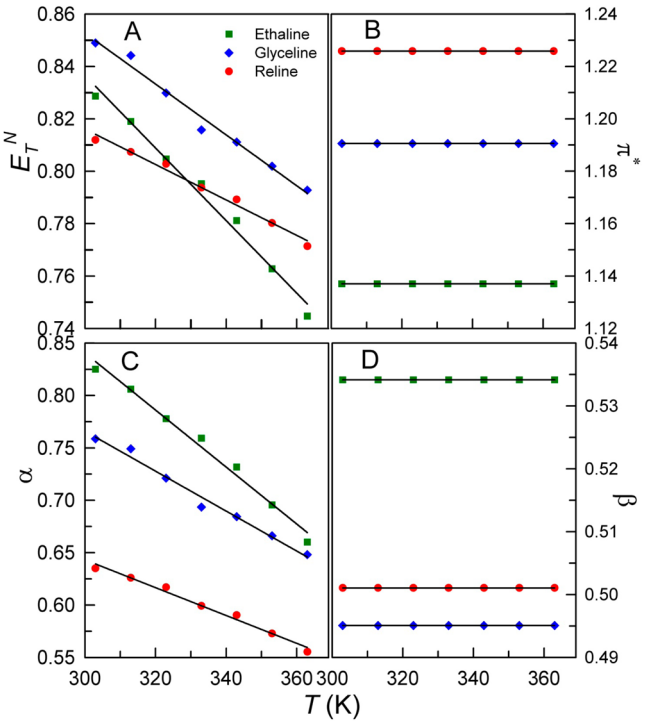


Figure 1. Variation of E_T^N (panel A), π^* (panel B), α (panel C), and β (panel D) with temperature in the three DESs—ethaline (■), glyceline (◆), and reline (●). Solid straight lines are the best fit obtained from the linear regression analysis. Errors in E_T^N, π^*, α, and β are $\leq \pm 0.01$.

polar liquids due partly to the destruction of the cooperative effect. For example, the static dielectric constant of water is observed to decrease as the temperature is increased to 100 °C.[34] It is important to mention that the observed decrease in E_T^N with increasing temperature within the DESs is similar to that observed within common and popular ionic liquids by Trivedi et al.,[35] Baker et al.,[27] and the El Seoud group[17,24] using different UV–vis absorbance probes.

Linear regression analysis of E_T^N versus temperature for ethaline, glyceline, and reline, respectively, reveals the decrease in E_T^N with increasing temperature to be linear in nature (Figure 1A). The temperature dependence of dipolarity/polarizability and/or HBD acidity as revealed by the betaine dye 33 is represented by the slope of the E_T^N versus temperature best fit straight line (Table 1). Among the three DESs, the absolute value of this slope decreases in the order ethaline > glyceline > reline, highlighting the ethaline to be the most sensitive DES as far as the response of E_T^N toward change in temperature is concerned. It appears that the dipolarity/polarizability and/or HBD acidity of a DES constituted of an H-bond donor with –OH functionalities is more sensitive to the change in temperature as compared to a DES constituted of an amide-based H-bond donor. Remarkably, the sensitivity of E_T^N toward the change in temperature within ethaline and glyceline, respectively, is found to be even higher than that within water, though the change in dipolarity/polarizability and/or HBD acidity as reflected via E_T^N with a change in temperature within reline is less than that observed for water. It is noteworthy that E_T^N variations with temperature within the DESs ethaline and glyceline are more dramatic than those reported within the common ionic liquids [bmim][PF_6] and [bmim][BF_4], respectively, and the slope of E_T^N versus temperature for reline is comparable to those for [bmim][PF_6] and [bmim][BF_4].[35]

Kamlet–Taft Parameters. In order to assess the temperature dependence of dipolarity/polarizability, HBD acidity, and HBA basicity separately of the DESs, we used well-documented empirical procedure for the estimation of Kamlet–Taft parameters using UV–vis molecular absorbance probes DENA and NA (Scheme 2).[36−38] We measured the wavelength of electronic absorbance maxima of the two probes, respectively, within each of the DESs in the temperature range 30–90 °C, and by combining them with $E_T(30)$, we obtained Kamlet–Taft empirical parameters π^* (dipolarity/polarizability), α (HBD acidity), and β (HBA basicity).[36−38] It is important to mention at this point that π^*, α, and β for the three DESs assessed by us at ambient temperature are in good agreement with those reported by other groups in the recent past.[10] Measured π^*, α, and β within ethaline, glyceline, and reline, respectively, at different temperatures are presented in Figure 1. A careful examination of the data reveals that, surprisingly, π^* and β of the three DESs do not change with temperature. This is in contrast to that reported for ionic liquids [bmim][PF_6] and [bmim][BF_4], respectively, where both π^* and β were found to decrease with increasing temperature.[35] While the π^* of water also does not change with a change in the temperature, the β for water is found to increase marginally with increasing temperature.[35] The parameter α on the other hand does decrease with an increase in temperature within all three DESs (the decrease in α with an increase in temperature can also be considered linear with slope recovered from the linear regression analysis within each of the DESs that is reported in Table 1). Similar to the trend for E_T^N, the sensitivity of α to temperature also follows the trend ethaline >

Table 1. Slopes Recovered from Linear Regression Analysis of E_T^N, π^*, α, β, Py I_1/I_3, ANS λ_{max}^{fluo}, and PRODAN λ_{max}^{fluo}, Respectively, versus Temperature for Three DESs[a]

	probe response						
	E_T^N ($\times 10^{-4}$ K^{-1})	π^* ($\times 10^{-4}$ K^{-1})	α ($\times 10^{-4}$ K^{-1})	β ($\times 10^{-4}$ K^{-1})	Py I_1/I_3 ($\times 10^{-4}$ K^{-1})	ANS λ_{max}^{fluo} ($\times 10^{-2}$ nm K^{-1})	PRODAN λ_{max}^{fluo} ($\times 10^{-2}$ nm K^{-1})
ethaline	−13.8 (±0.7)	∼0	−27.2 (±1.4)	∼0	−52.5 (±7.6)	∼0	∼0
glyceline	−9.7 (±0.5)	∼0	−19.1 (±1.1)	∼0	−59.3 (±6.5)	∼0	∼0
reline	−6.8 (±0.4)	∼0	−13.3 (±0.7)	∼0	−91.0 (±8.4)	∼0	∼0
water[b]	−9.1 (±0.5)	−0.6 (±0.0$_2$)	−17.2 (±1.7)	5.6 (±2.9)	−29.9 (±0.3)	NA[c]	NA[c]
[bmim][PF_6][b]	−8.5 (±0.4)	−8.4 (±0.8)	−9.3 (±0.9)	−5.2 (±1.4)	−42.4 (±0.7)	NA[c]	NA[c]
[bmim][BF_4][b]	−7.6 (±0.8)	−9.3 (±0.7)	−6.6 (±1.3)	−5.3 (±1.3)	−42.1 (±0.3)	NA[c]	NA[c]

[a]Corresponding slopes for water and ionic liquids [bmim][PF_6] and [bmim][BF_4] are also included for comparison. [b]From ref 35. [c]NA: not available.

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glyceline > reline. HBD acidity, α, decreases with increasing temperature for water and ionic liquids [bmim][PF_6] and [bmim][BF_4], respectively, as well.[27,35,39] It is noteworthy that, while the sensitivity of α with temperature within the three DESs is similar to that in water, it is significantly higher than those observed within common ionic liquids [bmim][PF_6] and [bmim][BF_4].[27]

The static dielectric constant of water is known to decrease with increasing temperature;[34] however, it is not manifested in the response of DENA (i.e., through parameter π^*). This may be attributed to the compensatory contributions from polarizability, although it is proposed that polarizabilities are only very slightly temperature dependent.[34] The behavior of DESs used is similar to that exhibited by water, and it is contrary to what was observed for ionic liquids [bmim][PF_6] and [bmim][BF_4]. It is inferred that the decrease in E_T^N with increasing temperature within DESs is due to the decrease in HBD acidity and not due to the decrease in dipolarity/polarizability of the medium. Apparently, within DESs, the increased average thermal reorientation of the dipoles leading to the destruction of the cooperative effect resulting in decreased dielectric constant with increasing temperature is perhaps compensated by the increase in polarizability of the medium. H-bonding is known to get weaker as the temperature is increased. It may also be highlighted that DESs constituted of HBDs having alcohol groups are more sensitive to temperature change than those possessing amide functionalities. As all three DESs contain the same ionic compound choline chloride, the decrease in HBD acidity of a DES with increasing temperature can be conveniently linked with the HBD acidity of the H-bond donor that is used to prepare the DES.

Response of Fluorescence Probes Pyrene, PRODAN, and ANS. Molecular fluorescence from an appropriate fluorophore is well-suited to furnish information regarding complex systems owing to the higher sensitivity and orthogonality of information inherent to fluorescence-based techniques.[40,41] In order to assess the effect of temperature on the DESs as manifested through the response of solvatochromic probes, we have selected three common but structurally diverse fluorescence polarity probes, pyrene, PRODAN, and ANS (Scheme 2).

Pyrene is one of the most widely used neutral fluorescence probes for polarity studies.[42,43] The pyrene solvent polarity scale (Py I_1/I_3) is defined by its I_1/I_3 emission intensity ratio, where I_1 is the intensity of the solvent-sensitive band arising from the $S_1(v = 0) \rightarrow S_0(v = 0)$ transition and I_3 corresponds to the solvent-insensitive $S_1(v = 0) \rightarrow S_0(v = 1)$ transition.[43] The I_1/I_3 ratio increases with increasing solvent dipolarity and is a function of both the solvent dielectric (ε) and the refractive index (n) via the dielectric cross term, $f(\varepsilon, n^2)$.[43] Experimentally obtained Py I_1/I_3 as a function of temperature in the range 30−90 °C within ethaline, glyceline, and reline, respectively, is depicted in Figure 2. It is interesting to note that, similar to E_T^N (and α), Py I_1/I_3 also decreases with an increase in temperature within all three DESs and the decrease can be considered linear in nature (Table 1). However, while the decrease of E_T^N and α with increasing temperature was more pronounced within ethaline followed by glyceline and was least within reline (*vide supra*), for Py I_1/I_3, the decrease follows the reverse trend: reline > glyceline > ethaline. It is easily conceivable as decrease in E_T^N with increasing temperature is due in major part to the decrease in HBD acidity, whereas, in the case of Py I_1/I_3, it is due to the decrease in solvent dielectric (ε) and/or refractive index (n). It is important to note that the decrease in Py I_1/I_3 with increasing temperature within the three DESs is more dramatic than those reported within ionic liquids, [bmim][PF_6] and [bmim][BF_4],[27,35] water,[35] 1-octanol, DMSO, propylene carbonate, butyl acetate, and dibutyl ether.[44] In general, the decrease in Py I_1/I_3 with increasing temperature is more pronounced in polar solvents including ionic liquids, and less pronounced in nonpolar solvents. As opposed to polar solvents, the inherently low ε (and dipole moment of the molecules) of the nonpolar solvents perhaps results in less dramatic changes in Py I_1/I_3 upon an increase in temperature. In this context, as reported earlier also,[45] the three DESs exhibit unusually high polar character as solubilizing media. Further, the dipolarity of the choline chloride-based DESs may change significantly with the change in the temperature.

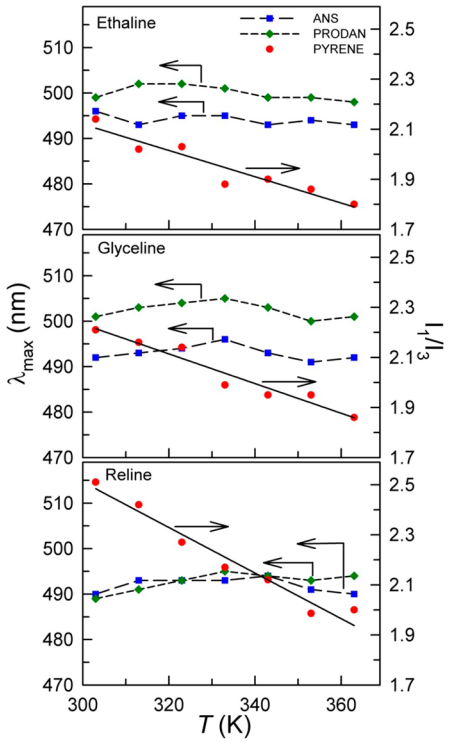


Figure 2. Variation of pyrene (Py, 1 μM) I_1/I_3 (●), 1-anilino-8-naphthalenesulfonate (ANS, 10 μM) λ_{max}^{fluo} (■), and 6-propionyl-2-(dimethylaminonaphthalene) (PRODAN, 10 μM) λ_{max}^{fluo} (◆) with temperature in the three DESs ($\lambda_{excitation}$ = 337, 346, and 350 nm for pyrene, ANS, and PRODAN, respectively, and excitation and emission slits are 2/2 nm). Solid straight lines are the best fit obtained from the linear regression analysis. The error in Py I_1/I_3 is $\leq \pm 0.02$, and the errors in PRODAN and ANS λ_{max}^{fluo} are $\leq \pm 2$ nm.

The responses (i.e., lowest energy fluorescence emission maxima, λ_{max}^{fluo}) of PRODAN, neutral, and ANS, negatively charged, photoinduced charge-transfer fluorescence probes (Scheme 2) within the DESs, on the contrary, do not change appreciably with the change in the temperature (Figure 2 and Table 1). In this respect, they behave more like UV−vis absorbance probes DENA and NA (π^* and β within DESs also do not change much with temperature, *vide supra*). It appears the responses of these probes are dominated by the HBA basicity of the milieu more than by other factors. It is clear that responses of PRODAN and ANS when dissolved in choline chloride-based DESs are not sensitive to the changes in the temperature of the system.

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Effect of Water. Choline chloride-based DESs are known to be hygroscopic in nature.[46] Many common and popular ionic liquids, such as alkylimidazolium ionic liquids with PF_6^- and Tf_2N^- anions, are also hygroscopic; however, they exhibit significantly restricted water miscibility.[27,47,48] On the contrary, choline chloride-based DESs used in this investigation are completely water miscible. As a result, water (an obvious environmentally benign substance) may be used as a cosolvent or an additive to modify properties of choline chloride-based DESs in an effective and favorable manner. In this regard, reports on physical properties (e.g., density, viscosity, and refractive index) of various (DES + water) mixtures have started to appear in the current literature.[46] It is imperative in this context to understand how the addition of water to DES affects the solvation behavior of a solute. Solvatochromic probes of different structure and functionalities can reveal key insights into the effect of added water on molecular solvation within a DES. We have also attempted to correlate the observations from the solvatochromic probe behavior with those acquired using noninvasive techniques, specifically FTIR absorbance and Raman spectroscopies.

E_T^N and Kamlet–Taft Parameters. Figure 3 presents experimentally obtained E_T^N, π^*, α, and β within aqueous mixtures of ethaline, glyceline, and reline, respectively, along with the ideal additive values represented by solid lines under ambient conditions. A cursory glance of the E_T^N implies almost a linear increase in E_T^N with the increase in the mole fraction of water (x_w) within the mixtures; however, a careful examination of the data reveals a slight preferential solvation of the betaine dye by the DESs within the aqueous mixture for all three DESs. Structural similarities between zwitterionic betaine dye and the quarternary ammonium cation of choline chloride constituting the DES may be responsible for this slight preferential solvation of the dye by the DES. Betaine dye was observed to be preferentially solvated by the ionic liquid within an aqueous mixture of a water-miscible ionic liquid [bmim][BF_4] as well; however, the extent of preferential solvation was fairly significant.[49,50] This was tentatively attributed to the similarity in the aromatic character associated with betaine dye and [$bmim^+$] of the ionic liquid.

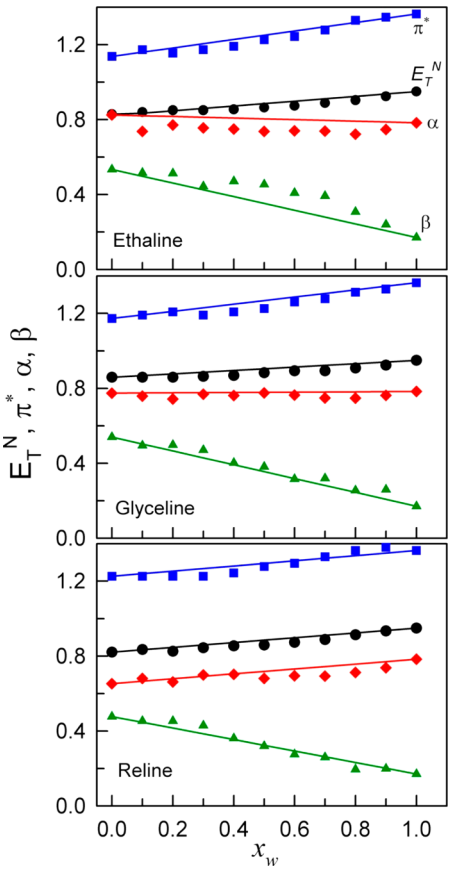


Figure 3. Variation of E_T^N (●), π^* (■), α (◆), and β (▲) with mole fraction of water (x_w) within aqueous DES mixtures under ambient conditions. Solid lines represent ideal-additive values. Errors in E_T^N, π^*, α, and β are $\leq \pm 0.01$.

The increase in dipolarity/polarizability, π^*, with increasing x_w within ethaline, glyceline, and reline, respectively, can also be considered linear in nature (Figure 3). The experimental π^* values within water-added DES, thus, are close to the values predicted from the ideal-additive behavior. This is in contrast to what was observed for the aqueous mixture of ionic liquid [bmim][BF_4] where preferential solvation of the probe DENA by the ionic liquid was clearly suggested.[50] The hint of slight preferential solvation of the betaine dye by the DES within the (DES + water) mixture could be attributed to the values of HBD acidity, α, within the mixture that appear to be somewhat lower than that predicted from ideal-additive behavior at several compositions (Figure 3). This is similar in trend, if not in magnitude, to what was observed for ([bmim][BF_4] + water) mixtures.[51] This somewhat lowered HBD acidity of the (DES + water) mixtures could be due to interspecies H-bonding between the DES and water as opposed to intraspecies H-bonding among water or DES molecules, respectively. As far as HBA basicity, β, is concerned, the values do appear to be closer to those predicted from ideal-additive behavior (Figure 3), though a hint of preferential solvation of the probe NA by the DES is suggested within (ethaline + water) mixtures. It is worth mentioning that β values of the aqueous mixture of the ionic liquid [bmim][BF_4] are significantly higher than those predicted from ideal-additive behavior, implying considerable preferential solvation of the probe NA by the ionic liquid.[49,51]

Fluorescence Probe Behavior. Responses of the three fluorescence solvatochromic probes, PRODAN (λ_{max}^{fluo}), ANS (λ_{max}^{fluo}), and pyrene (I_1/I_3), were recorded as water is added to three DESs, respectively (Figure 4). PRODAN is a neutral charge-transfer fluorescence probe. Experimental PRODAN λ_{max}^{fluo} are more or less similar to those predicted from ideal-additive behavior. This is in accord with the response of another neutral fluorescence probe, pyrene-1-carboxaldehyde, when dissolved in aqueous mixtures of ionic liquid [bmim][BF_4].[35] It appears the neutral fluorescence probes based on wavelength shift, similar to the neutral absorbance probes of wavelength shifts (DENA and NA), are not preferentially solvated by the DES within (DES + water) mixtures to any appreciable extent. The response of the anionic charge-transfer wavelength shift-based fluorescence probe, ANS, on the other hand, does exhibit preferential solvation by the DES within (ethaline + water) and (glyceline + water) mixtures, respectively (Figure 4). However, it is important to note that, within an aqueous mixtures of reline, ANS λ_{max}^{fluo} are not too different from those predicted from ideal-additive behavior. It appears H-bond donors ethylene glycol and glycerol that possess −OH groups may play a role in preferential interaction with the anionic probe ANS by the DES. ANS response is

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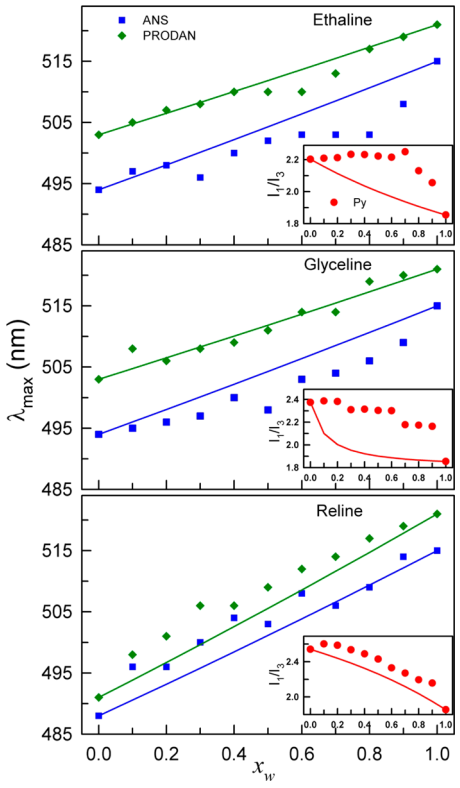


Figure 4. Variation of 1-anilino-8-naphthalenesulfonate (ANS, 10 μM) λ_{max}^{fluo} (■) and 6-propionyl-2-(dimethylaminonaphthalene) (PRODAN, 10 μM) λ_{max}^{fluo} (◆) with the mole fraction of water (x_w) within aqueous DES mixtures under ambient conditions. Insets show variation of pyrene (Py, 1 μM) I_1/I_3 (●) with mole fraction of water (x_w) under ambient conditions. Solid curves represent ideal-additive behavior. The error in Py I_1/I_3 is $\leq \pm 0.02$, and the errors in PRODAN and ANS λ_{max}^{fluo} are $\leq \pm 2$ nm.

known to get affected by the H-bonding capabilities of the cybotactic region more than the neutral probes.[52,53]

It is reported that Py I_1/I_3 values within an aqueous mixture of ionic liquid [bmim][BF_4] are significantly higher than those predicted from ideal-additive behavior.[35] This was interpreted in terms of the presence of strong solute−solvent or solvent−solvent interactions within the system. Within the three (DES + water) mixtures, Py I_1/I_3 are also observed to be significantly higher than those predicted from ideal-additive behavior (insets of Figure 4). Ideal additive Py I_1/I_3 values were calculated on the basis of the procedure outlined by Acree and co-workers.[54] As the presence of solvent−solvent interactions might have reflected in the behavior of other probes also, preferential solvation of the probe pyrene by the DES within (DES + water) mixtures may not be ignored here. It is possible that the aromatic π-cloud of pyrene, a large planar polycyclic aromatic hydrocarbon, preferentially interacts with the quarternary ammonium cation of the choline chloride. This is in agreement with the betaine dye response discussed earlier.

The solvatochromic probe responses from DESs as water is added are suggestive of preferential solvation of the solutes by the DES within an aqueous DES mixture. It appears the response of pyrene, a neutral polycyclic aromatic hydrocarbon fluorescence probe, is one of the most sensitive probes toward the presence of water in the DES. Pyrene response is affected by the static dielectric constant and/or the refractive index of the medium. Although preferential solvation of pyrene by the DES within the aqueous DES mixture may result in a pyrene response which is closer to the pyrene response in neat DES, the presence of solvent−solvent interactions between DES and water may not be completely ruled out. As one piece of evidence, the Py I_1/I_3 values, which are higher in DES than in water, are anomalously high in water-added DES mixtures. Surprisingly, in the DES-rich regime, the Py I_1/I_3 values become even higher than the Py I_1/I_3 values observed in neat reline. This is observed in the DES-rich region for all three aqueous mixtures of DESs. This unusual phenomenon, termed "hyperpolarity" in recent literature especially in context with ionic liquid mixtures,[55] cannot be explained on the basis of preferential solvation. Solvent−solvent interactions with or without solute−solvent interactions (giving rise to preferential solvation) must be evoked. Further, it is also interesting to note that the deviation of experimental Py I_1/I_3 from ideal-additive Py I_1/I_3 is the most prominent for the (glyceline + water) mixture followed by the (ethaline + water) mixture (insets of Figure 4). Although "hyperpolarity" in Py I_1/I_3 is observed for the (reline + water) mixture also in the reline-rich regime, the deviation of experimental Py I_1/I_3 from the ideal-additive Py I_1/I_3 is the least for the aqueous mixtures of reline among the three (DES + water) systems investigated.

FTIR Absorbance and Raman Spectroscopy. We have recently reported the temperature dependence of the bulk properties, density and dynamic viscosity, of aqueous mixtures of glyceline and reline, respectively.[18,19] We found that excess molar volumes (V^E) were negative for both (glyceline + water) and (reline + water) mixtures at all compositions in the temperature range from 20 to 90 °C. One of the interesting features was that the maximum absolute value of $V^E = -0.3372$ cm^3 mol^{-1} ($x_w \sim 0.6$) at 20 °C for an aqueous mixture of glyceline was more than 2 times the maximum absolute value of $V^E = -0.1522$ cm^3 mol^{-1} ($x_w \sim 0.7$) for an aqueous mixture of reline at the same temperature. The difference in excess logarithmic viscosities, $(\ln \eta)^E$, for aqueous mixtures of reline as opposed to that of glyceline was also pointed out in these investigations.[18] Specifically, while $(\ln \eta)^E$ were positive for aqueous mixtures of glyceline at all compositions, for (reline + water) mixtures, $(\ln \eta)^E$ were mostly negative. While H-bonding within a mixture between the components forming the mixture usually leads to positive $(\ln \eta)^E$ as the mixture viscosity comes out to be higher than that predicted, interstitial accommodation of one component with the other within the mixture may lead to negative $(\ln \eta)^E$ as, in this case, the mixture viscosity would be less than that expected. Both of these factors, however, would lead to negative V^E. It was proposed that, while interstitial accommodation of water within reline was perhaps the major factor governing interactions within (reline + water) mixtures, the H-bonding between water and the components of glyceline (i.e., glycerol and/or choline chloride) was the major interaction present within (glyceline + water) mixtures. This is easily attributed to the more efficient H-bonding between water and glycerol as compared to that between water and urea within the aqueous DES mixture as glycerol possesses three alkyl —OH groups as opposed to two —NH_2 groups on the C═O functionality in urea. This would lead to increased contraction in volume within aqueous mixtures of glyceline as opposed to that within aqueous mixtures of reline.

The outcomes of the aforementioned bulk property measurements are in good agreement with those obtained from solvatochromic probe responses. Specifically, interstitial accommodation of water within the network of the

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components forming the DES along with interspecies H-bonding between water and DES (or its components) appear to feature prominently in governing the interactions present within aqueous DES mixtures. In order to further explore the interactions present within (DES + water) mixtures, we have used spectroscopic methods where aqueous mixtures of DESs are investigated in a noninvasive manner. Specifically, we have acquired FTIR absorbance and Raman spectra of aqueous mixtures of reline and glyceline, respectively, under ambient conditions (Figures 5 and 6).

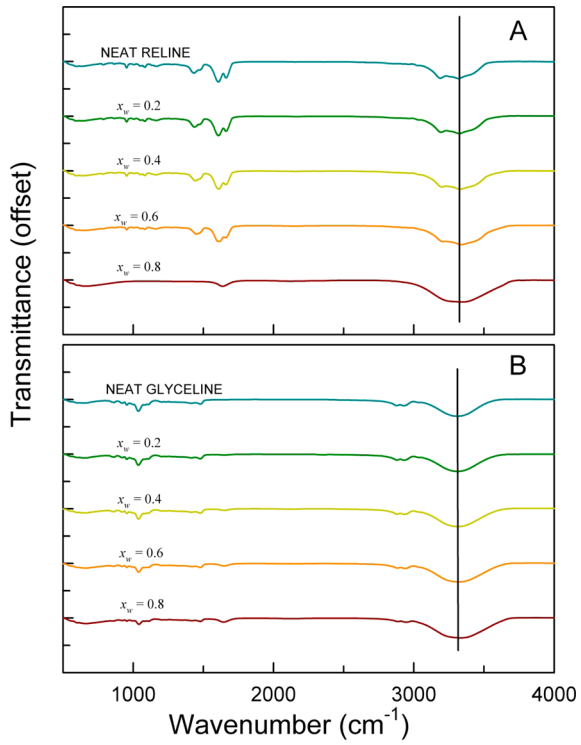


Figure 5. FTIR absorbance spectra of (reline + water) mixture (panel A) and (glyceline + water) mixture (panel B) under ambient conditions.

The FTIR absorbance spectrum of reline under ambient conditions is highlighted, among others, by ν_s NH_2 at 3317 cm^{-1}, δ_s NH_2 at 3189 cm^{-1}, δ_{as} NH_2 at 1606 cm^{-1}, ν_{as} CN at 1165 cm^{-1}, and ν_{as} CCO at 953 cm^{-1} peaks. Our FTIR spectrum of reline is in good agreement with that reported in the literature.[56] Figure 5A presents FTIR spectra of reline as water is added. A careful examination of the spectra reveals no appreciable shifts in peak positions for ν_{as} CN and ν_{as} CCO as water is added. However, all FTIR absorbance peaks pertaining to $-NH_2$ are shifted hypsochromically to some extent (ν_s NH_2 blue shifts to 3353 cm^{-1} and δ_{as} NH_2 to 1626 cm^{-1}). It is clear that urea−urea and urea−chloride H-bonding is diminished within the (reline + water) mixture upon addition of water. Further, we do not have clear evidence of very strong interspecies H-bonding between water and the components of reline (i.e., urea and/or choline chloride). This is in agreement with our solvatochromic probe responses where the HBD acidity of the aqueous reline mixture comes out to be either close to or slightly lower than that predicted by the ideal-additive response.

It has been hypothesized earlier that, when reline is formed, the H-bonding between urea molecules and that between

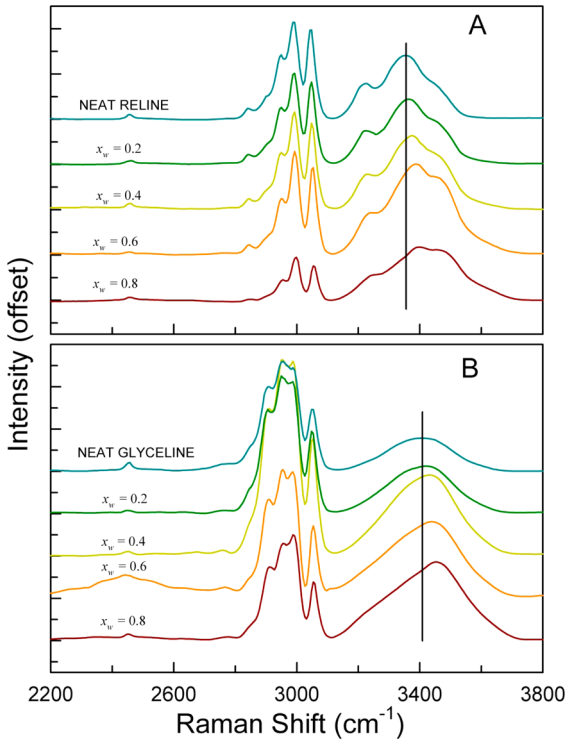


Figure 6. Raman spectra ($\lambda_{excitation}$ = 532 nm) of (reline + water) mixture (panel A) and (glyceline + water) mixture (panel B) under ambient conditions.

choline and chloride decrease and, in turn, the H-bonding between urea and chloride increases significantly.[57,58] The observation that, in reline, the anion interacts with urea more than the cation was reconfirmed very recently.[46] On the basis of MD simulations, it is shown that, as water is added to reline, while urea−urea and urea−chloride H-bonding decreases, the urea−choline H-bonding decrease is not as significant. Addition of water definitely helps form interspecies H-bonding, however, the overall H-bonding interactions do not alter much within reline as water is added. We propose interstitial accommodation of water within the molecular network of reline is perhaps the dominant interaction present within this system. Density and viscosity data of the aqueous reline mixture also support this proposition (*vide supra*). FTIR absorbance spectra of glyceline in the presence of water show the −OH stretch of glyceline becomes similar than that of water (Figure 5B), suggesting relatively stronger H-bonding to be present within the system. This is in agreement with the density and viscosity data of the aqueous glyceline mixture.[18]

The Raman spectra of reline as water is added also support the lack of the presence of extensive interspecies H-bonding within the mixture (Figure 6A). A careful examination of the Raman spectra of the aqueous reline mixture reveals absence of any appreciable shifts in any Raman band of reline upon addition of water. It is also clear that the region representing H-bonding interactions involving reline (the highest energy band) does not undergo significant changes as water is added to reline. This again implies the H-bonding within reline to be similar in the absence or the presence of water; as revealed by the data, the interstitial accommodation of water molecules within the reline molecular framework would not give rise to much change in Raman spectra of reline. The Raman spectrum of glyceline undergoes relatively more changes as water is

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added to glyceline (Figure 6B). Though the peaks corresponding to choline chloride do not show appreciable shifts, the ones corresponding to glycerol do exhibit certain changes as water is added to glyceline. As water is added, the combined peak representing all −OH functionalities present within the system does shift more than that expected (6, 7, 8, and 11 cm^{-1} more than that expected for x_w = 0.2, 0.4, 0.6, and 0.8, respectively). This hints at the presence of H-bonding interactions between the −OH functionalities within glyceline and added water molecules. It appears the interspecies H-bonding with water primarily involves glycerol −OH as opposed to choline −OH as the ratio of the peaks corresponding to CH_2 Raman vibrational modes of glycerol also change considerably (Figure 6B). The Raman spectral analysis also suggests the (glyceline + water) mixture to have relatively more interspecies H-bonding as compared to the (reline + water) mixture, where interstitial accommodation of water within the reline molecular network appears to dominate.

CONCLUSIONS

Responses of solvatochromic probes dissolved in three choline chloride-based DESs reveal that important physicochemical properties of DESs can be effectively modulated by changing temperature or adding water to the DES. Increasing temperature results in considerably decreased H-bond donating acidity of the DESs; dipolarity/polarizability and H-bond-accepting basicity do not change with temperature. The overall decrease in the polarity of the DESs with an increase in temperature is revealed by the response of the fluorescence probe pyrene. Responses from different solvatochromic probes along with outcomes from FTIR absorbance and Raman spectroscopic measurements reveal the H-bonding interactions between added water and DESs to be more pertinent for ethaline and glyceline, and less for reline. While interspecies H-bonding appears to be important within aqueous mixtures of ethaline and glyceline, interstitial accommodation of water within the H-bonded network of reline appears to dominate within aqueous mixtures of this DES. The structural differences between H-bond donors are proposed to be the reason for these observations. 1,2-Ethanediol and glycerol possessing two and three −OH groups, respectively, on an otherwise saturated hydrocarbon skeleton have relatively stronger H-bonding capabilities as compared to urea which contains two $-NH_2$ groups joined by a carbonyl functionality. The interactions present within DESs and their aqueous mixtures are amply revealed by these investigations. The outcomes of these studies may help establish choline chloride-based DESs and their aqueous mixtures as inexpensive environmentally benign solubilizing media in chemical sciences.

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Notes

The authors declare no competing financial interest.

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REFERENCES

(1) McGavock, W. G.; Bryant, J. M.; Wendlandt, W. W. Urea Complexes of Lithium Chloride. *Science* **1956**, *123*, 897.
(2) Abbott, A. P.; Capper, G.; Davies, D. L.; Rasheed, R. K.; Tambyrajah, V. Novel Solvent Properties of Choline Chloride/Urea Mixtures. *Chem. Commun.* **2003**, 70−71.
(3) Zhang, Q.; Vigier, K. D. O.; Royer, S.; Jerome, F. Deep Eutectic Solvents: Syntheses, Properties and Applications. *Chem. Soc. Rev.* **2012**, *41*, 7108−7146.
(4) Abbott, A. P.; Boothby, D.; Capper, G.; Davies, D. L.; Rasheed, R. K. Deep Eutectic Solvents Formed Between Choline Chloride and Carboxylic Acids: Versatile Alternatives to Ionic Liquids. *J. Am. Chem. Soc.* **2004**, *126*, 9142−9147.
(5) Abbott, A. P.; Cullis, P. M.; Gibson, M. J.; Harris, R. C.; Raven, E. Extraction of Glycerol from Biodiesel into a Eutectic Based Ionic Liquid. *Green Chem.* **2007**, *9*, 868−872.
(6) Abbott, A. P.; Harris, R. C.; Ryder, K. S.; D'Agostino, C.; Gladden, L. F.; Mantle, M. D. Glycerol Eutectics as Sustainable Solvent Systems. *Green Chem.* **2011**, *13*, 82−90.
(7) Gorke, T.; Srienc, F.; Kazlauskas, R. J. Hydrolase-catalyzed Biotransformations in Deep Eutectic Solvents. *Chem. Commun.* **2008**, 1235−1237.
(8) Weaver, K. D.; Kim, H. J.; Sun, J.; MacFarlane, D. R.; Elliott, G. D. Cyto-toxicity and Biocompatibility of a Family of Choline Phosphate Ionic Liquids Designed for Pharmaceutical Applications. *Green Chem.* **2010**, *12*, 507−513.
(9) Radosevic, K.; Bubalo, M. C.; Srcek, V. G.; Grgas, D.; Dragičevic, T. L.; Redovnikovic, I. R. Evaluation of Toxicity and Biodegradability of Choline Chloride Based Deep Eutectic Solvents. *Ecotoxicol. Environ. Saf.* **2015**, *112*, 46−53.
(10) Harris, R. M. Physical Properties of Alcohol Based Deep Eutectic Solvents. Doctoral Thesis, University of Leicester, 2008.
(11) Leron, R. B.; Li, M. H. High-Pressure Density Measurements for Choline Chloride: Urea Deep Eutectic Solvent and its Aqueous Mixtures at T = (298.15 to 323.15) K and up to 50 MPa. *J. Chem. Thermodyn.* **2012**, *54*, 293−301.
(12) Esquembre, R.; Sanz, J. M.; Wall, J. G.; del Monte, F.; Mateo, C. R.; Ferrer, M. L. Thermal Unfolding and Refolding of Lysozyme in Deep Eutectic Solvents and Their Aqueous Dilutions. *Phys. Chem. Chem. Phys.* **2013**, *15*, 11248−11256.
(13) Leron, R. B.; Wong, D. S. H.; Li, M. H. Densities of a Deep Eutectic Solvent Based on Choline Chloride and Glycerol and its Aqueous Mixtures at Elevated Pressures. *Fluid Phase Equilib.* **2012**, *335*, 32−38.
(14) Siongco, K. R.; Leron, R. B.; Li, M. H. Densities, Refractive Indices, and Viscosities of N,N-diethylethanol Ammonium Chloride−Glycerol or −Ethylene Glycol Deep Eutectic Solvents and Their Aqueous Solutions. *J. Chem. Thermodyn.* **2013**, *65*, 65−72.
(15) Siongco, K. R.; Leron, R. B.; Caparanga, A. R.; Li, M. H. Molar Heat Capacities and Electrical Conductivities of Two Ammonium-Based Deep Eutectic Solvents and Their Aqueous Solutions. *Thermochim. Acta* **2013**, *566*, 50−56.
(16) Wu, S. H.; Caparanga, A. R.; Leron, R. B.; Li, M. H. Vapor Pressure of Aqueous Choline-based Deep Eutectic Solvents (Ethaline, Glyceline, Maline, and Reline) at 30−70 °C. *Thermochim. Acta* **2012**, *544*, 1−5.
(17) Lin, C. M.; Leron, R. B.; Caparanga, A. R.; Li, M. H. Henry's Constant of Carbon dioxide-Aqueous Deep Eutectic Solvent (Choline Chloride/Ethylene Glycol, Choline Chloride/Glycerol, Choline Chloride/Malonic Acid) Systems. *J. Chem. Thermodyn.* **2014**, *68*, 216−220.
(18) Yadav, A.; Trivedi, S.; Rai, R.; Pandey, S. Densities and Dynamic Viscosities of (Choline Chloride + Glycerol) Deep Eutectic Solvent and its Aqueous Mixtures in the Temperature Range (283.15 to 363.15) K. *Fluid Phase Equilib.* **2014**, *367*, 135−142.

(19) Yadav, A.; Pandey, S. Densities and Viscosities of (Choline Chloride + Urea) Deep Eutectic Solvent and its Aqueous Mixtures in the Temperature Range 293.15 to 363.15 K. *J. Chem. Eng. Data* **2014**, *7*, 2221−2229.
(20) Reichardt, C. Solvatochromism, Thermochromism, Piezochromism, Halochromism, and Chiro-Solvatochromism of Pyridinium N-Phenoxide Betaine Dyes. *Chem. Soc. Rev.* **1992**, *21*, 147−153.
(21) Nishida, S.; Morita, Y.; Fukui, K.; Sato, K.; Shiomi, D.; Takui, T.; Nakasuji, K. Spin Transfer and Solvato-/Thermochromism Induced by Intramolecular Electron Transfer in a purely Organic Open-Shell System. *Angew. Chem., Int. Ed.* **2005**, *44*, 7277−7280.
(22) Nicolet, P.; Laurence, C. Polarity and Basicity of Solvents. Part 1. A Thermosolvatochromic Comparison Method. *J. Chem. Soc., Perkin Trans.* **1986**, *2*, 1071−1079.
(23) Linpo, Y.; Chen, G. Z. Cryo-solvatochromism in Ionic Liquids. *RSC Adv.* **2014**, *4*, 40281−40285.
(24) Sato, B. M.; de Oliveira, C. G.; Martins, C. T.; El Seoud, O. A. Thermo-solvatochromism in Binary Mixtures of Water and Ionic Liquids: On the Relative Importance of Solvophobic Interactions. *Phys. Chem. Chem. Phys.* **2010**, *12*, 1764−1771.
(25) Wei, X.; Yu, L.; Wang, D.; Chen, G. Z. Thermo-solvatochromism of Chloro-Nickel Complexes in 1-hydroxyalkyl-3-methyl-imidazolium Cation Based Ionic Iiquids. *Green Chem.* **2008**, *10*, 296−305.
(26) Khupse, N. D.; Kumar, A. Contrasting Thermosolvatochromic Trends in Pyridinium-, Pyrrolidinium-, and Phosphonium-Based Ionic Liquids. *J. Phys. Chem. B* **2010**, *114*, 376−381.
(27) Baker, S. N.; Baker, G. A.; Bright, F. V. Temperature-Dependent Microscopic Solvent Properties of 'Dry' and 'Wet' 1-Butyl-3-methylimidazolium Hexafluorophosphate: Correlation with $E_T(30)$ and Kamlet−Taft Polarity Scales. *Green Chem.* **2002**, *4*, 165−169.
(28) Reichardt, C. Solvatochromic Dyes as Solvent Polarity Indicators. *Chem. Rev.* **1994**, *94*, 2319−2358.
(29) Reichardt, C. Pyridinium N-Phenolate Betaine Dyes as Empirical Indicators of Solvent Polarity: Some New Findings. *Pure Appl. Chem.* **2004**, *76*, 1903−1919.
(30) Muldoon, M. J.; Gordon, C. M.; Dunkin, I. R. Investigations of Solvent−Solute Interactions in Room Temperature Ionic Liquids Using Solvatochromic Dyes. *J. Chem. Soc., Perkin Trans.* **2001**, *2*, 433−435.
(31) Sarkar, A.; Trivedi, S.; Pandey, S. Unusual Solvatochromism within 1-Butyl-3-methylimidazolium Hexafluorophosphate + Poly(ethylene glycol) Mixtures. *J. Phys. Chem. B* **2008**, *112*, 9042−9049.
(32) Reichardt, C. *Solvents and Solvent Effects in Organic Chemistry*, 3rd ed.; Wiley-VCH: Weinheim, Germany, 2003.
(33) Marcus, Y. *Introduction to Liquid State Chemistry*; Wiley-Interscience: New York, 1977.
(34) Lide, D. R., Ed. *CRC Handbook of Chemistry and Physics*, 87th ed.; CRC Press: Boca Raton, FL, 2006.
(35) Trivedi, S.; Malek, N. I.; Behera, K.; Pandey, S. Temperature-Dependent Solvatochromic Probe Behavior within Ionic Liquids and (Ionic Liquid + Water) Mixtures. *J. Phys. Chem. B* **2010**, *114*, 8118−8125.
(36) Kamlet, M. J.; Abboud, J. L.; Taft, R. W. The Solvatochromic Comparison Method. 6. The π^* Scale of Solvent Polarities. *J. Am. Chem. Soc.* **1977**, *99*, 6027−6038.
(37) Taft, R. W.; Kamlet, M. J. The Solvatochromic Comparison Method. 2. The α-Scale of Solvent Hydrogen-Bond Donor (HBD) Acidities. *J. Am. Chem. Soc.* **1976**, *98*, 2886−2894.
(38) Kamlet, M. J.; Taft, R. W. The Solvatochromic Comparison Method. I. The β Scale of Solvent Hydrogen-Bond Acceptor (HBA) Basicities. *J. Am. Chem. Soc.* **1976**, *98*, 377−383.
(39) Cammarata, L.; Kazarian, S. G.; Salter, P. A.; Welton, T. Molecular States of Water in Room Temperature Ionic Liquids. *Phys. Chem. Chem. Phys.* **2001**, *3*, 5192−5200.
(40) Suppan, P.; Ghoneim, N. *Solvatochromism*; RSC Publishing: Cambridge, U.K., 1997.
(41) Valeur, B.; Berberan-Santos, M. N. *Molecular Fluorescence Principles and Applications*, 2nd ed.; Wiley-VCH: Weinheim, Germany, 2012.
(42) Street, K. W., Jr; Acree, W. E., Jr. Experimental Artifacts and Determination of Accurate Py Values. *Analyst* **1986**, *111*, 1197−1201.
(43) Karpovich, D. S.; Blanchard, G. J. Relating the Polarity-Dependent Fluorescence Response of Pyrene to Vibronic Coupling. Achieving a Fundamental Understanding of the Py Polarity Scale. *J. Phys. Chem.* **1995**, *99*, 3951−3958.
(44) Rai, R.; Pandey, S. Solvatochromic Probe Response within Ionic Liquids and their Equimolar Mixtures with Tetraethylene Glycol. *J. Phys. Chem. B* **2014**, *118*, 11259−11270.
(45) Pandey, A.; Rai, R.; Pal, M.; Pandey, S. How Polar are Choline Chloride based Deep Eutectic Solvents? *Phys. Chem. Chem. Phys.* **2014**, *16*, 1559−1568.
(46) Shah, D.; Mjali, F. S. Effect of Water on the Thermo-physical properties of Reline: An Experimental and Molecular Simulation Based Approach. *Phys. Chem. Chem. Phys.* **2014**, *16*, 23900−23907.
(47) Fletcher, K. A.; Pandey, S. Effect of Water on the Solvatochromic Probe Behavior within Room-Temperature Ionic Liquid 1-Butyl-3-methylimidazolium Hexafluorophosphate. *Appl. Spectrosc.* **2002**, *56*, 266−271.
(48) Baker, S. N.; Baker, G. A.; Munson, C. A.; Chen, F.; Bukowski, E. J.; Cartwright, A. N.; Bright, F. V. Effects of Solubilized Water on the Relaxation Dynamics Surrounding 6-Propionyl-2-(N,N-dimethylamino)naphthalene Dissolved in 1-Butyl-3-methylimidazolium Hexafluorophosphate at 298 K. *Ind. Eng. Chem. Res.* **2003**, *42*, 6457−6463.
(49) Sarkar, A.; Ali, M.; Baker, G. A.; Tetin, S. Y.; Ruan, Q.; Pandey, S. Multiprobe Spectroscopic Investigation of Molecular-level Behavior within Aqueous 1-Butyl-3-Methylimidazolium Tetrafluoroborate. *J. Phys. Chem. B* **2009**, *113*, 3088−3098.
(50) Ali, M.; Sarkar, A.; Tariq, M.; Ali, A.; Pandey, S. Dilute Aqueous 1-Butyl-3-methylimidazolium Hexafluorophosphate: Properties and Solvatochromic Probe Behavior. *Green Chem.* **2007**, *9*, 1252−1258.
(51) Sarkar, A.; Pandey, S. Solvatochromic Absorbance Probe Behavior and Preferential Solvation in Aqueous 1-Butyl-3-methylimidazolium Tetrafluoroborate. *J. Chem. Eng. Data* **2006**, *51*, 2051−2055.
(52) Stryer, L. Fluorescence Spectroscopy of Proteins. *Science* **1968**, *162*, 526−533.
(53) Kosower, E. M.; Dodiuk, H.; Tanizawa, K.; Ottolenghi, M.; Orbach, N. Intramolecular Donor-Acceptor Systems. Radiative and Nonradiative Processes for the Excited States of 2-N-Arylamino-6-naphthalenesulfonate. *J. Am. Chem. Soc.* **1975**, *97*, 2167−2178.
(54) Acree, W. E., Jr.; Wilkins, D. C.; Tucker, S. A.; Griffin, J. M.; Powell, J. R. Spectrochemical Investigations of Preferential Solvation. 2. Compatibility of Thermodynamic Models versus Spectrofluorometric Probe Methods for Tautomeric Solutes Dissolved in Binary Mixtures. *J. Phys. Chem.* **1994**, *98*, 2537−2544.
(55) Sarkar, A.; Trivedi, S.; Baker, G. A.; Pandey, S. Multiprobe Spectroscopic Evidence for "Hyperpolarity" within 1-Butyl-3-methylimidazolium Hexafluorophosphate Mixtures with Tetraethylene Glycol. *J. Phys. Chem. B* **2008**, *112*, 14927−14936.
(56) Yue, D.; Jia, Y.; Yao, Y.; Sun, J.; Jing, Y. Structure and Electrochemical Behavior of Ionic Liquid Analogue Based on Choline Chloride and Urea. *Electrochim. Acta* **2012**, *65*, 30−36.
(57) Perkins, S. L.; Painter, P.; Colina, C. M. Molecular Dynamic Simulations and Vibrational Analysis of an Ionic Liquid Analogue. *J. Phys. Chem. B* **2013**, *117*, 10250−10260.
(58) Perkins, S. L.; Painter, P.; Colina, C. M. Experimental and Computational Studies of Choline Chloride-Based Deep Eutectic Solvents. *J. Chem. Eng. Data* **2014**, *59*, 3652−3662.